



OHIO RIVER TRAIL FEASIBILITY STUDY

June 17, 2010

Ohio River Trail Proposed Route - Phasing Strategy

Realizing that the improvements identified in this plan collectively represents approximately \$5M to \$6M in new infrastructure investments in 2010 dollars, it is important to consider how individual projects can be organized to make the overall implementation of the project manageable. An important consideration when developing the phasing strategy is the desire to maximize overall connectivity along the trail route as quickly as possible. The key to this strategy is the utilization of the existing Pennsylvania Bicycle Route A as the current bicycle and pedestrian spine. This existing on-road route allows for individual projects to occur while providing a connector route for the overall trail corridor. This also allows for the accommodation of the disconnected trail segments as portions of the proposed ORT Trail route are constructed in the locations where it differs from the Pennsylvania Bicycle Route A.

The following break-down represents an itemized list of projects, and in some cases sub projects, into separate phase to be achieve within an approximately 10 to 15 year completion horizon.

Phase 1 - On-Road Segment Improvements and Signing: Years 1 and 2

The projects in this phase represent the segments of the proposed ORT Trail route that will consist of on-road segments, either on the current Pennsylvania Bicycle Route A or via other existing roadways. In most cases the improvements required for the initial implementation of these segments will consists of trailblazer signing for the route and enhanced safety improvements in the form of signing, roadway striping and markings and crossing improvements, as well as pedestrian/bicyclist signals, warning lighting and traffic calming. Some of these segments are identified to ultimately have new trail facilities constructed parallel to the existing roadway, specifically along Woodlawn Road within Segments 200 and 300. With the goal of establishing the route and the identity of the ORT Trail as soon possible, most of these segments can be initially addressed as on-road routes. In most cases any improvements to these on-road segments would immediately serve the local community as well as connect to Pennsylvania Bicycle Route A.

There is one proposed construction oriented project in this phase, which is Segment 600.6 in Coraopolis which runs through the core of the Borough's Downtown Revitalization Area and is described in more detail below.

The segments included in the Phase 1 are:

Segment 100 - Monaca Borough

100.1 - Monaca Rochester Bridge to Monaca Memorial Field via 9th Street, Washington Avenue and 16th Street. Improvements along this on-road segment will consist of trailblazer signing that will begin to establish the entity of the ORT Trail and establish its connection to the North Shore of the Ohio River, via the Monaca Rochester Bridge.

Segment 200 – City of Aliquippa

200.4 - Woodlawn Road from 1st Street to 6th Street (approximately the western point of the proposed BCED roadway improvement project). In Phase 4 this segment is targeted for a 10' wide paved trail of asphalt to be located in the wide grass median between Woodlawn Road and River Avenue. In Phase 1 this segment could be signed as a trailblazer route and could serve the adjacent neighborhood.

Segment 300 – City of Aliquippa

300.2 - Woodlawn Road from the Entrance to USG Plant (approximately the eastern point of the proposed BCED roadway improvement project) to a Point Just East of Steel Street (western terminus). This segment is part of the portion of Woodlawn Road that has been improved. In Phase 4 it is targeted for a 10' wide gravel side path but could be signed as an on road route prior to trail construction.

300.3 Woodlawn Road in the Vicinity of the Franklin Avenue Underpass. This segment is part of the Woodlawn Road that has been improved. In Phase 4 it is targeted for a 10' wide asphalt side path but could be signed as an on road route.

300.4 - Woodlawn Road from a Point Just East of Steel Street (western end) to Steel Street (eastern terminus). This segment is part of the Woodlawn Road that has been improved. In Phase 4 it is targeted for a 10' wide gravel side path but could be signed as an on road route in this phase.

Segment 400 – South Heights Borough

400.5 - Route 51 from Power Plant Road to Crescent Township Line. This segment of Route 51 is designated as Pennsylvania Bicycle Route A. It is targeted for a 10' wide asphalt side path along the frontage of the former power plant property but could be signed as an on road route prior to trail construction. Recent crosswalk improvements were made in this segment of Route 51. Trailblazer signing and additional safety improvements, such as traffic calming, could be considered as well as curb bumps outs for key crossing locations and enhanced roadway striping and markings.

Segment 400 - Crescent Township

400.6 - Route 51 from South Heights Borough Line to Jeannette Street. This segment of Route 51 is designated as Pennsylvania Bicycle Route A. It is targeted for a 10' wide asphalt side path along the frontage of the former power plant property but could be signed as an on road route prior to trail construction. Recent crosswalk improvements were made in this segment of Route 51. Trailblazer signing and additional safety improvements, such as traffic calming, could be considered as well as curb bumps outs for key crossing locations and enhanced roadway striping and markings.

400.7 - Route 51 from Jeanette Street to Main Street, McCutcheon Way and Riverview Park. This segment of Route 51 is designated as Pennsylvania Bicycle Route A. Trailblazer signing and additional safety improvements could be considered including curb bumps outs at key route crossing locations. This area serves as a major gateway through the Township with residential and commercial uses and should be considered for streetscape and safety improvements including traffic calming, curb bump-outs, crossing warning signals and architectural lighting and landscaping.

400.9 - Dashields Lock Road from McCutcheon Way to the Crescent Township Line and U.S. Army Corps of Engineers Parking Lot. This area experiences very little traffic since it is exclusively a roadway to access the Dashields Lock Complex. This area will only require on-road trailblazing signing.

Segment 600 – Coraopolis Borough

600.4 - Kendall Street from Birch Avenue to 1st Avenue. This segment of the trail travels through a transitional area between the industrial zone to the west and the primarily residential/mixed-use area to the east. This segment is targeted as an on-road route and will only require trailblazing signing.

600.5 - 1st Avenue from Kendall Street to Broadway Street and Broadway Street to Pine Alley. 1st Avenue is the main residential street that is closest and parallel to the Ohio River in Coraopolis. It is targeted as an on-road route and will only require trailblazing signing and possibly some traffic calming since some vehicular speeding was witnessed during field observations. Broadway Street is a residential street and will only require on-road trailblazer signing.

600.6 – Pine Alley from Broadway Street to Montour Street. Pine Alley is targeted to be a “green alleyway” which will include safety improvements including striping and lighting as well as improved streetscaping and site specific stormwater management improvements to promote better urban stormwater runoff management. The on-road route through this segment could be initially signed with trailblazer signing, however, this segment also represents an excellent route for the core of Coraopolis to access the pending Montour Junction Recreation Complex located at the east end of the Borough.

600.7 - Montour Street from 4th Avenue to the Montour Junction Access Point Adjacent to the CSX Railroad Mainline. This small segment is a critical link between the core downtown area of the Borough

that will be served by Segment 600.6 and the Montour Junction property. This route should be signed once the Montour Junction property is publicly accessible. The area is targeted as an on-road route, however, additional safety improvements should be considered for the intersection of 4th Avenue and Montour Street. These improvements could include curb bump outs, enhanced crosswalk striping and crossing warning signals.

Phase 2 – Linkages to On-Road Segments and the Montour Trail: Years 3 through 5

The projects in this phase represent the first phase of major trail segment construction. The determination of which segments to include in this phase is a function of their ability to expand on Phase 1 on-road segments or to take advantage of planned construction activities along portions of the proposed ORT Trail route.

The segments included in the Phase 2 are:

Segment 100 - Monaca Borough

100.2 - Memorial Field Alleyway from Indiana Avenue at 16th Street, Underneath the East Monaca Rochester Bridge to the Alleyway at the Eastern Terminus of Indiana Avenue. This segment is proposed to consist of a 10' wide asphalt path. The improvements in this area will require grading and retention wall improvements for the portion along the football field fence. A major portion of the existing alleyway consists of old brick pavers which could also be reused in their entirety or integrated into the design of the paving surface with an asphalt trail section. This segment will also require stormwater management improvements in the form of a bio-swale along the field fence and the removal of invasive plant species on the upper slope that are currently compounding soil erosion issues on the slope. At the point where the trail turns to travel underneath the bridge, a retaining wall "bench" will need to be created to accommodate the trail, adjacent to the bridge abutment. Additional public amenities such as wooden riverfront overlooks are proposed since this is a very attractive location along the riverfront and would provide a public enhancement to the community.

100.3 – The Alleyway at the Eastern Terminus of Indiana Avenue to Moor Industrial Park Property Line. The improvements to this segment would consist of reconstructing the 20' wide alleyway to accommodate on-road routing of the trail. This will also allow for a complete loop from Indiana Avenue at 16th Street to the terminus of Indiana Avenue on the east side of 17th Street.

100.5 – Industrial Park Road from Pennsylvania Avenue/Route 51 to the Monaca Borough/Center Township Line. This segment is included in Phase 2 since there is currently a pending proposal to locate a new riverfront refinery operation on the BET-TECH property located at the Monaca Borough/Center Township Municipal Line. In order to accommodate the truck traffic generated by this proposed facility, a new roadway, or major roadway improvements are being considered for Industrial Park Road. This roadway is currently privately owned and, if improved, would extend from the Pennsylvania Avenue/Route 51 Bridge to Monaca Borough/Center Township Municipal Line. The proposed trail

improvements for this segment would consist of signing and safety improvements for a small portion of the roadway adjacent to the industrial building located at the intersection with Pennsylvania Avenue/Route 51. Starting at the point of the old cemetery, as a new 10' wide gravel side path will be creating between Industrial Park Road and the CSX railyard right-of-way, extending to the Monaca Borough/Center Township Municipal Line.

Segment 200 – City of Aliquippa

200.5 - Woodlawn Road from 6th Street (approximately the western point of the proposed BCED roadway improvement project) to Segment 300 divider line (approximate midpoint of BCED property). This segment is targeted for a 10' wide gravel side path to be located on the side of the roadway closest to the railroad. This segment is included in Phase 2 based on the fact that BCED is actively pursuing funding to construct a new roadway in this location and if additional engineering and construction funding can be secured for trail design and specifications, the trail and the roadway could potentially be constructed concurrently.

Segment 300 – City of Aliquippa

300.1 - Woodlawn Road from the Segment 300 Divider Line (approximate midpoint of BCED property) to the Existing USG Complex Access Roadway. This is a continuation of Segment 200.5 and is targeted for a 10' wide gravel side path to be located on the side of the roadway closest to the railroad. This segment is included in Phase 2 based on the fact that BCED is actively pursuing funding to construct a new roadway in this location and if additional engineering and construction funding can be secured for trail design and specifications, the trail and the roadway could potentially be constructed concurrently.

300.5 – Woodlawn Road from Steel Street to the Eastern Property Line of the BCED/BET-TECH Tin Mill Property. This segment is part of the portion of Woodlawn Road that has not yet been improved and is schedule for phasing after Segments 200.5 and 300.1. It is proposed for a 10' wide gravel side path on the railroad side of the roadway. Funding to include the engineering for the trail in the final roadway engineering drawings and specifications should be pursued to ensure that the trail and the roadway can be constructed concurrently.

300.6 Woodlawn Road from the Eastern Property Line of the BCED/BET-TECH Tin Mill Property to the Segment 400 Divider Line. The ownership of this segment of Woodlawn is currently in contention. Based on parcel information and verbally provided information, it appears that the roadway is currently privately owned. Acquiring this roadway as public right-of-way would be the first step needed to create a trail in this location. Today it could not even be designated as an on-road route without an access easement. This segment is targeted for a 10' wide gravel path on the railroad side of the roadway.

Segment 400 – Hopewell Township

400.1 - Woodlawn Road from the Segment 400 Divider Line to the Hopewell Township/South Heights Borough Municipal Line. As with Segment 300.6, the ownership of this segment of Woodlawn is currently in contention. Based on parcel information and verbally provided information, it appears that the roadway is currently privately owned. Acquiring this roadway as public right-of-way would be the first step needed to create a trail in this location. Today it could not even be designated as an on-road route without an access easement. This segment is targeted for a 10' wide gravel path on the railroad side of the roadway.

Segment 400 – South Heights Borough

400.2 – Woodlawn Road from Hopewell Township/South Heights Borough Municipal Line to North Street Underpass. As with Segment 300.6 and 400.1, the ownership of this segment of Woodlawn is currently in contention. Based on parcel information and verbally provided information, it appears that the roadway is currently privately owned. Acquiring this roadway as public right-of-way would be the first step needed to create a trail in this location. Today it could not even be designated as an on-road route without an access easement. This segment is targeted for a 10' wide gravel path on the railroad side of the roadway.

400.3 - Off Road Multi-Use Path From the North Street CSX Underpass to the Power Plant Road Bridge. This segment of the trail will consist of a 10' wide gravel off-road multi use trail. It travels through a series of parcels that are owned by CSX or Reliant Energy. The trail will diverge from North Street near the CSX underpass and continue parallel to the river along what is believed to be an abandoned rail siding that once served the now closed power plant. The trail will continue along this alignment until it reaches the Power Plant Road Bridge over the CSX railroad.

400.4 – Power Plant Road from the Bridge to Route 51. Power Plant Road is currently a private roadway. An access easement or outright purchase of the right-of-way would need to occur in order to route the trail along this roadway. This segment will include the bridge span as well as the roadway. The trail along this segment would be an on-road route, including on the bridge, and will only require trailblazing signing.

Segment 400 - Crescent Township

400.8 – McCutcheon Way from Main Street to Dashields Lock Road. This small segment consists of a relatively low-trafficked roadway. The roadway, however, has significant horizontal and vertical constraints and poor site lines. The roadway is a horseshoe curve, switch back, that rises from the low elevation of Riverview Park to the significantly higher elevation of Shouse Park. Based on field observations it appears that a side path could be created along with a switch-back boardwalk to address current constraints. More detailed engineering studies will need to be performed to determine what design options are feasible.

Segment 500 – Moon Township

500.3 - Western Boundary of Parcel West of Sewickley Bridge to American Bridge Way. This segment represents a key segment of the trail within the overall ORT trail route. Moon Township has targeted a former brownfield property located at Valley Ambulance Drive and the river for a new waterfront park. The property is currently undergoing environmental remediation through PA Act 2 and is owned by the Moon Township Municipal Authority. The Authority also owns, or is in the process of acquiring, several other parcels to the west of the proposed park parcel. Township has several water supply wells located on these parcels. The proposed trail would extend from the western parcel boundary, parallel to the river and the CSX railroad right-of-way to the proposed park parcel. The trail is proposed to be a 10' wide gravel path. A portion of the trail through the proposed park property would be located on the concrete building slab from the former building, where a railroad siding once entered the industrial facility. The trail is proposed to be asphalt in the location of the concrete slab to the at-grade railroad crossing at Valley Ambulance Drive.

The trail will cross the at-grade crossing and then travel parallel to the CSX railroad on a utility corridor right-of-way from Valley Ambulance Drive to 4th Avenue, just west of Thorn Run. A bridge span or culvert will be required to cross Thorn Run, behind the PennDOT guiderail in this location.

The railroad right-of-way for the entire length of the trail in Segment 500.3 is proposed to be fenced.

Segment 600 – Coraopolis Borough

600.1 - American Bridge Way from 4th Avenue to 3rd Avenue Right-of-Way. This short section of recently reconstructed roadway is wide and was designed to accommodate heavy tanker trailer truck traffic generated by the Petroleum Products gasoline distribution plant located on American Bridge Way. As a result of this truck traffic an asphalt or concrete side path is proposed for the eastern side of American Bridge Way. The creation of this trail segment will require widening the existing sidewalk in this location, which is landscaped area with American Bridge Corp's facility sign.

600.2 - 3rd Avenue (Paper Street) from American Bridge Way to Watt Street. This area, which aligns with 3rd Avenue in other sections of the Borough, appears to be public right-of-way based on the location of the existing fence line along American Bridge's property. Digital tax parcel information does not show this as a separate parcel, however. Instead, it appears to be split between American Bridge and CSX. Additional property ownership research will be required to determine actual ownership. This area is targeted for a 10' wide asphalt path with right-of-way fencing along the CSX railroad.

600.3 – Watt Street from 3rd Avenue to Birch Avenue. This segment is proposed as an on-road route for the trail and would only require trailblazer signing. It is a short segment located between two segments that require full construction so it is not included as a Phase 1 segment.

600.4 – Birch Avenue from Watt Street to Kendall Street. Birch Avenue is currently in very poor condition and is only utilized as a rear service alley to several adjacent industrial properties. This segment is targeted for a 20' wide alleyway reconstruction with landscape and stormwater management improvements. Once reconstructed, the trail would be an on-road on this portion and would require trailblazer signing.

600.8 - Montour Street to Montour Trail Milepost "0". The exact routing of this trail segment will need to be determined as part of the site master planning effort for the overall Montour Junction Sportsplex Project. Depending on its context within the park it could either be gravel or a paved asphalt path with a width of 10' or 12' or greater. The segment is the critical link between Coraopolis's downtown core and the existing Montour Trail system.

Phase 3 – Key Infill Linkages: Years 5+

Segment 100 - Monaca Borough

100.4 – Moor Industrial Park Connector from Indiana Avenue to Pennsylvania Avenue/Route 51 and Industrial Park Road. This segment will require coordination with the owners of the Moor Industrial Park to create the trail connection between what should be two completed segments. The proposed trail in this segment will consist of a 10' wide trail of asphalt construction. If the trail is constructed while the existing industrial use remains on the site, the proposed trail will include landscaping and possibly some right-of-way fencing.

This parcel currently appears to be under-utilized and the portion that includes the proposed alignment appears to be vacant or only utilized for storage. The proposed trail alignment would travel as close to the western parcel boundary as possible. It has not been determined if the owners of the Moor Industrial Park will allow a 10' wide easement or outright purchase of right-of-way along the edge of the parcel on the east side of the bridge approach and through a portion of the property (between two apparently vacant buildings) to reach Pennsylvania Avenue. It is presumed that a trail could be incorporated into the redevelopment of this site. It is believed that the owners are currently considering redevelopment plans. Ideally, the requirement to accommodate and construct the public trail should be codified through modifications to ordinances and the adoption of the trail feasibility study.

Segment 200 – Center Township

200.1 - Future Woodlawn Road from the Monaca Borough/Center Township Municipal Line to Center Township/Hopewell Township Municipal Line. The majority of Segment 200 is controlled by BET-TECH and consists of the former J&L Steel Plant property. It is anticipated that a trail through the area will not be feasible until a new roadway, Future Woodlawn Road, is created. When that occurs a 10' wide gravel or asphalt side path (depending on the grade of the trail) should be created along the railroad side of the roadway. This trail location will greatly reduce any conflicts between truck traffic entering or leaving

the roadway and potentially crossing the trail, since in most cases the development will occur along the riverside of the roadway.

The probable cost of this segment assumes that a new roadway and trail can be accommodated above the existing outfall of Elkhorn Run. This area is very constrained as result of the location and configuration of the CSX track. This constraint is partly due to changes that CSX made to its track alignment in this area. According to information provided by Richard Dorothy of C.J. Betters Enterprises, the railroad legal right-of-way agreements in this area, which date back to J&L Steel, are very complicated and open to interpretation. More detailed engineering would be required to determine if a two-lane roadway along with a 10' wide sidepath could be created in this area. There is also an option to locate the trail route at the bottom of the slope, near the edge of the river in this location, but the feasibility of that alignment could only be determined once an engineered alignment between the proposed refinery facility and existing dock is determined. A crossing of the day-lighted portion of Elkhorn Run would be required if this route is chosen and the cost will likely be greater.

Segment 200 – Hopewell Township

200.2 - Future Woodlawn Road from Center Township/Hopewell Township Municipal Line to Existing Terminus of the Reconstructed Woodlawn Road (adjacent to Aliquippa Water Authority Plant). The majority of Segment 200 is controlled by BET-TECH and consists of the former J&L Steel Plant property. It is anticipated that a trail through the area will not be feasible until a new roadway, Future Woodlawn Road, is created. When that occurs, a 10' wide gravel or asphalt side path (depending on the grade of the trail) should be created along the railroad side of the roadway. This location will greatly reduce any conflicts between truck traffic entering or leaving the roadway and potentially cross the trail, since in most cases the development will occur along the riverside of the roadway.

Segment 200 – City of Aliquippa

200.3 - Woodlawn Road at the Water Authority Plant to 1st Street. This segment is ultimately targeted for 10' wide gravel side path but could be signed as an on-road route to serve the nearby neighborhood of Aliquippa. Woodlawn Road in this area has already been improved so the side path would need to be added to the railroad side of the roadway. The need for a side path along this segment will be more critical as industrial development occurs on the BET-TECH property which will likely generate more significant truck traffic.

Segment 500 – Moon Township

500.1 – U.S. Army Corps of Engineers Dashields Lock Area. The creation of a trail along this segment, when combined with Segment 500.2 would create more than 2.5 continuous miles of dedicated multi-use trail in very close proximity to the river's edge and also would connect two major riverfront public parks; Shouse Park in Crescent Township and the proposed waterfront park in Moon Township. This trail segment would consist of a 10' wide multi-use path. The path would be primarily gravel, although along

a small portion was the trail where it travels up in elevation from Dashields Lock Road to a height that allows it to pass behind the Lock House, it will need to be asphalt. In order to create the trail behind the lock complex and “bench” will need to be created by the placement of a series of parallel retaining walls.

500.2 – Stoops Ferry Area. This area is the most topographically challenging portion of the entire proposed ORT Trail route. In order to locate the trail in this location and achieve the 50’ setback from the track centerline as required by CSX, a boardwalk structure will be required. It is believed that one parcel; at the location of the former P&LE Stoops Ferry station will require acquisition from CSX. This entire segment will likely require special permitting due to its special construction techniques, earth disturbance in close proximity to the floodway and the need to access the area via barge for construction.

Phase 4 – Strategic Upgrading of On-Road Segments with Side Paths: Years 5+

Segment 200 – City of Aliquippa

200.4 - Woodlawn Road from 1st Street to 6th Street (approximately the western point of the proposed BCED roadway improvement project). This phase will consist of constructing a 10’ wide asphalt or concrete trail to be located in a wide grass median between Woodlawn Road and River Avenue. This segment’s location is an area that is park or boulevard-like due to the amount of lawn area and the orientation of the parallel roadways. During field observations, numerous residents were observed walking and biking along this stretch of un-utilized roadway. As a result, this portion of the trail could be treated like a promenade with architectural lighting, an allée of street trees and other site amenities to provide broader community enhancement to area of Aliquippa that has been severely impacted by the demise of the J&L Steel Plant.

Segment 300 – City of Aliquippa

300.2 - Woodlawn Road from the Entrance to the USG Plant (approximately the eastern point of the proposed BCED roadway improvement project) to a Point Just East of Steel Street (western terminus). This segment is part of the Woodlawn Road that has been improved. It is targeted for a 10’ wide gravel side path that will be located on the railroad side of the roadway. The portion of trail that is directly adjacent to the CSX railroad will be fenced.

300.3 Woodlawn Road in the Vicinity of Franklin Avenue Underpass. This trail segment is part of the portion of Woodlawn Road that has been improved. It is targeted for a 10’ wide asphalt side path. An asphalt trail is required in this area due to the grades that exist to approach the underpass from the westerly direction. This area will require more detailed engineering to determine an exact route of side path. The intersection with Franklin Avenue and the circulation constraints of the underpass have the potential to create traffic conflicts between trucks and bicyclists. Ideally the trail would be located on the river side of the roadway for this segment, which will require a crossing of Woodlawn Road at each end of the segment to align with the adjoining segments.

300.4 - Woodlawn Road from a Point Just East of Steel Street (western end) to Steel Street (eastern terminus). This segment is part of the portion of Woodlawn Road that has been improved. It is targeted for a 10' wide gravel side path located along the railroad side of the roadway.

Segment 400 – South Heights Borough

400.5 - Route 51 from Power Plant Road to Crescent Township Line. Currently there is a large front setback along the Reliant Power Plant property. If this site remains in its existing condition, a 10' wide side path could be created along the Route 51 frontage of the parcel, with additional streetscaping, buffer plantings and landscaping. If the property is redeveloped, the incorporation of 10' wide multi-use path should be required as part of any redevelopment site plan approval process. This requirement should be required if the property is rezoned in any manner.

Segment 400 - Crescent Township

400.6 - Route 51 from South Heights Borough Line to Jeannette Street. As with Segment 400.5 a 10' wide side path along the front of the property should be created or routed through the property, should any redevelopment occur on the site.